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RESEARCH ARTICLE

Alpha and theta oscillations in mental addition for high and low performers

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Abstract

While many contemporary studies aim to explore the sources of notable individual differences in arithmetic skills, this study specifically aims to highlight cognitive differences between high and low math performers. Thirty-six undergraduate female students who were identified as either high or low math performers, according to an arithmetic fluency test, were recruited. In the main experiment, EEG recordings were taken, while the participants performed a mental addition task. The mental addition problems were classified as either easy or difficult, and were presented to participants in several forms. The results indicated that problem difficulty increases the gap in accuracy attainment between high and low math performers. Additionally, high performers displayed larger alpha power during mental arithmetic in P7, corresponding to the left parietal lobe. This indicated that combining behavioral and neural data can improve our understanding of the differences between high and low math performers. Interpretations and implications are discussed.

Keywords Brain oscillation · Math abilities · Mental arithmetic · Solving strategy · Visuospatial working memory

Introduction

Math abilities are an essential element of IQ, predicting academic and professional achievement (see Neisser et al. 1996 for a review). Research in numerical cognition is persistent in discovering the factors related to arithmetic skills, forming basic math ability (Imbo and LeFevre 2009; 2010; Lee and Kang 2002; Logie et al. 1994; Otsuka and Osaka, 2015), and underlying neural correlated activities (Artemenko et al. 2018; De Smedt et al. 2009; Grabner et al. 2007; 2009; Menon et al. 2000b; Micheloyannis et al. 2005; Proverbio and Carminati 2019).

Based on an extensive review of literature on math abilities, Dowker (2005) concluded that individual differences in arithmetic abilities are largely attributed to flexibility and appropriate strategy choices in arithmetic performance. Suitable strategies vary according to task demands (Campbell and Xue 2001; Otsuka and Osaka 2015) and other cognitive abilities (Campbell and Xue 2001; Imbo and LeFevre 2009; 2010; Otsuka and Osaka 2015). Several studies have aimed to examine the role of solving strategies (Campbell and Xue 2001; Cragg et al. 2017; Hubber et al. 2013; Rosenberg-Lee et al. 2009; Tschentscher and Hauk 2014) and working memory components (Cavdaroglu and Knops 2015; Clearman et al. 2017; Imbo and LeFevre 2010; Lee and Kang 2002; Logie et al. 1994; Otsuka and Osaka 2015; Seitz and Schumann-Hengsteler 2000) in variant settings of mental arithmetic. This study follows this approach by examining the differences between high and low math performers in different settings of mental addition, by comparing performance accuracy and oscillatory brain activities during mental addition.

Theoretical background

Early studies in neuroimaging have found an increase in activities in the left prefrontal, frontal, and specific parietal regions, when performing arithmetic tasks (Micheloyannis et al. 2005; Rivera et al. 2005; Roland and Friberg 1985; Rueckert et al. 1996; Stanescu-Cosson et al. 2000).

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The role of the parietal circuits includes general domain abilities, such as retrieving verbal information in the left angular region, attention in the posterior superior parietal region, and perceiving mental number representation in the intraparietal region (Dehaene et al. 2003). Some activities in the frontal lobe have been found to correspond to working memory function (Micheloyannis et al. 2005; Sammer et al. 2007). Furthermore, the prefrontal region within the frontal area has been found to be largely responsible for activities related to planning, imagination, and strategy selection organisation (Roland and Friberg 1985; Taillan et al. 2015).

Some studies show that an increase of arithmetic skills leads to a shift from prefrontal and more general frontal activation to activation in two areas in the parietal region: the right intraparietal sulcus and left angular gyrus. This reflects the shift from use of general cognitive abilities, including planning, selecting strategies, and working memory, which are related to the frontal areas, to representation of numbers during calculation and arithmetic fact retrieval. Representation of numbers during calculation is related to the right intraparietal sulcus whereas arithmetic fact retrieval is related to the left angular gyrus (Delazer et al. 2005; Rivera et al. 2005).

Studies further specified the roles of certain parietal regions and revealed another shift within the parietal lobe from intraparietal sulcus to angular gyrus. Here, the shift occurs to be from representation and calculation of numbers to mere retrieval of arithmetic facts (Delazer et al. 2005; Prado et al. 2013).

However, conflicting results were found by methods of comparing studies investigating higher and lower math performers, showing inconsistencies in finding an increase of activity in angular gyrus during arithmetic performance in higher and in lower math performers. Grabner and colleagues (2007) showed larger angular gyrus activity in parietal area in higher math performers than that in lower performers, and suggested that this reflects different levels of dependence on retrieval strategies in mental arithmetic (Grabner et al. 2007). Menon and colleagues (2000a) found a different pattern of activation, whereby activities in the angular gyrus were lower during arithmetic performance in higher than in lower math performers. It should be noted that the participants in Menon's study were all high performers, who were divided based on the experimental arithmetic task into perfect (100% correct answers) and imperfect (above 90% correct answers). Additionally, the arithmetic tasks involved three operant addition and subtraction problems to encourage participants to engage in procedural strategies (e. g., deconstructing problems into smaller ones), rather than using retrieval strategies to find answers. Grabner and colleagues used single and double-digit multiplication problems, which are more likely to depend on answer retrieval, and the participants were intermediate in their math abilities (i.e., they were chosen based on a screening test from around the midpoint of normal distribution). This discrepancy between studies emphasizes the importance of considering task demands when comparing arithmetic accuracy and adopting strategies along with brain activities in high and low math performance.

In electrophysiological studies of mental arithmetic abilities, studies either focus on synchrony of brain oscillation, or their power when performing mental arithmetic (Artemenko et al. 2018; De Smedt et al. 2009; Proverbio and Carminati 2019). Synchrony of brain oscillation reflects coherence between brain regions while EEG power reflects the amount of activity in each frequency band. These terms are linked, as alpha synchronization is related to the inhibition and reduction of cortical activities, while the absence of alpha synchronization (alpha desynchronization) indicates cortical arousal (an increase in wave amplitude corresponding to the high demands of cognitive processes). In contrast, theta synchronization is a sign of cortical arousal, and the terms theta synchronization and theta power are positively related (Klimesch 1999; for an illustration see Soltanlou et al. 2019).

De Smedt and colleagues (2009) compared synchrony of alpha and theta when performing simple and complex addition and subtraction tasks. Simple tasks include problems without carrying or borrowing numbers, and the revealed answer is a small number, less than 10. Complex problems, in contrast, have integers with two digits and the answers are larger than 10, which can involve carrying or borrowing numbers. The results showed alpha desynchronization when participants solved complex problems, and theta synchronisation when participants solved simple problems. Nevertheless, synchrony of brain oscillation was not consistent across EEG studies of mental arithmetic. A study comparing undergraduate Chinese students who had demonstrated average performance in math achievement tests, with those designated as gifted students (i.e., had a high IQ, and had won an award in math), showed that theta synchronisation increases in the prefrontal lobe only when gifted undergraduates work on a verbal deductive reasoning task (Zhang et al. 2015).

Other studies comparing high and low math performers showed that brain activities when performing simple and complex tasks can vary according to participants' proficiency. Artemenko and colleagues (2018) compared performance of high and low performers on simple addition and subtraction, where the problems are solved with no carrying or borrowing, and complex problems, with carrying or borrowing. Their results showed that low performers were slower when responding to complex problems compared to when responding to simple problems, but their neural activities were not different between simple and complex problems. In contrast, high performers showed a larger increase in brain activities in the frontal region when solving complex

problems') compared to with simple problems. They argued that high performers need to recruit additional brain regions to solve complex problems while low performers recruited large brain activities for both simple and complex problems. This indicates that low performers, compared to high performers, are less efficient in performing both simple and complex tasks.

Proverbio and Carminati (2019) argued, based on similar findings, that high performers were more efficient in processing mental arithmetic on behavioral and neural levels. They revealed larger activation in the prefrontal area for high performers accomplishing complex tasks, compared to when completing simple problems. In contrast, low performers showed the same level of increase in EEG activities with both simple, medium, and complex problems. As neural correlates are relatively unspecified, research on behavioral performance in arithmetic, and testing performance of high and low performers under several task demands, would further disentangle the complex mechanisms related to math performance.

Explanations of brain activities during mental arithmetic are partially imported from working memory studies. Previous studies show that alpha power decreases with the increase of working memory load in several brain regions, including the prefrontal, frontocentral and posterior parietal regions (Borghini et al. 2014; Foxe and Snyder 2011; Klimesch 1999; Micheloyannis et al. 2005). In contrast, theta power increases with both working memory load and fact retrieval (Borghini et al. 2014; Klimesch et al. 2005; Raghavachari et al. 2006; Sammer et al. 2007). An increase in theta power spreads across the brain cortex, and is unrelated to the specific process occurring; it arises when cognitive processing takes place, regardless of which task is performed (Sammer et al. 2007). Alternatively, the spread of theta oscillations can be explained by the various strategies used to solve mental arithmetic tasks. Micheloyannis and colleagues (2005) showed the spread of brain activities when solving multi-digit multiplication tasks and explained their findings by suggesting the necessity of using a sequence of strategies to solve arithmetic problems. Moreover, using either retrieval or procedural strategies to perform mental arithmetic is related to specific parietal lobe regions (Grabner et al. 2009). This emphasises the importance of combining several parameters in one study to confirm previous findings on behavioral performance in mental arithmetic and its neural basis.

A precaution that should be taken when formulating any hypotheses regarding neural bases of mental arithmetic is that previous findings of the neural bases of arithmetic performance come either from fMRI studies (Ansari et al. 2011; Menon et al. 2000b; Prado et al. 2011) that rely on BOLD signals, or from EEG studies that rely on event-related or brain oscillatory components (De Smedt et al. 2009; Micheloyannis et al. 2005; Proverbio and Carminati 2019). These are differently related to behavioral performance (Winterer et al. 2007). Hence, it is not useful for the current study to adopt assumptions for EEG activities based solely on the results of fMRI studies.

Previous studies have employed several task demands to explore a participant's use or choice of solving strategies (Campbell and Xue 2001; Cragg et al. 2017; Hubber et al. 2013; Sidney et al. 2018). A common paradigm is the dual task paradigm, where participants perform mental arithmetic concurrently with a secondary task (Cavdaroglu and Knops 2015; Clearman et al. 2017; Cragg et al. 2017; Imbo and LeFevre 2010; Lee and Kang 2002; Logie et al. 1994; Otsuka and Osaka 2015; Seitz and Schumann-Hengsteler 2000). The rationale behind this paradigm is to load one component of working memory, including the phonological loop for verbal information, the visuospatial sketchpad for visual and spatial information, and executive control for controlling high demanding tasks (Baddeley 2000). If the loaded component is essential for mental arithmetic, the participant's performance suffers under that loading condition.

In this intensive work on the role of working memory component in simple and complex arithmetic operations, few studies focused on the visuospatial working memory component (Cavdaroglu and Knops 2015; Clearman et al. 2017; Cragg et al. 2017; Hubber et al. 2013). The role of visuospatial working memory component is controversial, as some studies found no interfering effect of visuospatial loading on mental arithmetic (Imbo and Vandierendonck 2007; Logie 1994), while other studies showed an impairing effect (Clearman et al. 2017; Cragg et al. 2017; Imbo and LeFevre 2010; Lee and Kang 2002). Another debate is related to the visuospatial component's role. Imbo and LeFevre (2010) reveal that reliance on visuospatial working memory when solving complex arithmetic is moderated by several factors, including the performance level and the method of presenting the problem (i.e., horizontally or vertically). In their study, performance by Chinese adult participants in complex subtraction was impaired by visuospatial loading when the problems were presented vertically, compared to when they were presented horizontally. One plausible explanation by Imbo and LeFevre (2010) for this relationship between vertical presentation and visuospatial working memory, is that vertical presentation encouraged decomposing the problems and numbers into smaller ones, and these procedures benefit from visuospatial working memory (see also Foley et al. 2016). This links the role of visuospatial working memory with how the problems are presented and suggest that the method of presentation might engage or exile the visuospatial working memory component when solving mental arithmetic.

Other findings support a view of equivalent interfering effect of visuospatial loading secondary tasks on mental

arithmetic in different settings, denying the moderating role of task demands (Cragg et al. 2017). In addition, the role of visuospatial working memory in applying procedural strategies to solve mental arithmetic was examined by Hoober and colleagues (2013), who compared the effect of a concurrent visuospatial secondary task on mental addition performed with one of three strategies, namely recall and two other procedural strategies: problem decomposition into smaller steps, or counting starting from one operant until the answer was revealed. Their results reveal the significant role of visuospatial working memory when using procedural strategies to solve addition problems. Cragg and colleagues (2017) showed that concurrent working memory tasks, including visuospatial working memory, impair mental arithmetic performance regardless of used strategies. In this realm, exploring the differences between high and low performers in mental arithmetic in their strategy choice—with or without the presence of a secondary visuospatial working memory task would clarify part of the argument and is crucial to understanding how high and low math performers differ in their mental arithmetic skills.

Overall, previous studies show visuospatial working memory to be an important resource in solving complex arithmetic problems (Imbo and LeFevre 2010; Zhou et al. 2006). Additionally, visuospatial working memory is related to procedural strategies when performing mental arithmetic (Hubber et al. 2013). Furthermore, high and low performers differ in their flexibility and strategy choice (Dowker 2005; Otsuka and Osaka 2015). Testing the effect of several task demands on participant accuracy and correlated EEG activities, including presentation method, loading of visuospatial working memory and complexity, and whether these changes in task demands have variant effects on higher and on lower math performance would be informative. Indeed, these implementations can partially attribute differences between high and low math performers in mental arithmetic to either differences in flexibility in strategy selection or in visuospatial working memory resources.

If high performers showed larger variation in strategies used, compared to low performers, according to the method of presentation, complexity or visuospatial loading, then the findings would give further evidence on the importance of flexibility in choosing a suitable strategy for a task. If low performers also showed flexibility in their use of solving strategies according to the method of presentation and complexity, but showed dampened performance and fixation to one strategy or mixed strategies within the same condition, then the argument would be about the limited working memory resources for low performers. If both strategies are fixed in all conditions, and visuospatial working memory loading dropped performance in all loaded conditions in low performers compared to high performers, then it could be argued that there would be differences between high and low math performers in both flexibility in choosing strategies and sources of working memory.

Aims and expectations

As the study aims to contribute to our understanding of the roots of differences between high and low math performers, it aims to compare accuracy and related alpha and theta activities when high and low math performers solve addition problems. It further aims to examine the role of visuospatial working memory in different task demands (i.e., simple vs. complex tasks and horizontal vs. vertical presentation). In all the above conditions, it examines to what extent high and low math performers depend on recall or procedural strategies during the abovementioned task parameters.

Significant differences between high and low math performers are expected in complex tasks but not in simple tasks. This is based on the findings of previous studies, where behavioral and neural differences between high and low math performers were shown in difficult problems (Proverbio and Carminati 2019). Regarding brain oscillations in the frontal (De Smedt et al. 2009; Micheloyannis et al. 2005) and parietal lobes (De Smedt et al. 2009), a decrease in alpha power is expected when solving difficult problems versus easy problems. Based on the findings in previous studies, changes in oscillation powers are expected to occur in these regions, which show neural changes in activities related to difficult mental arithmetic in these regions (Stanescu-Cosson et al. 2000; Menonet al. 2000b). Additionally, the decrease in frontal lobe alpha power is expected to be larger for high math performers in difficult arithmetic problems than for low math performers. This is derived from the findings that alpha power decrease is related to visual attention (Klimesch 1999) and central executives (Sauseng et al. 2005), which function more in high performers (Dowker 2005). In contrast, building expectations for whether theta power differs between high and low math performers during mental arithmetic is difficult because theta power increase is related to working memory loading (Klimesch 1999; Klimesch et al. 2005; Raghavachari et al. 2006; Sammer et al. 2007). It is unclear whether low performers fail to recruit working memory resources to calculate difficult problems, or whether they load their working memory like high performers do but fail because they lack other cognitive resources.

No clear predictions can be made relating to the role of visuospatial working memory and task demands. Loading on visuospatial working memory has a negative effect on complex arithmetic on participants from largely variant cultures (Western and Canadian) (Clearman et al. 2017; Cragg et al. 2017; Imbo and LeFevre 2010). Only Chinese participants showed a selective effect of visuospatial load on presentation method, showing the interference of vertically presented

subtraction problems (Imbo and LeFevre 2010). It is hard to build assumptions based on speculations about the role of visuospatial working memory in mental arithmetic, involving imagery of calculus or any form of visual representation in mental sketchpad (Dehaene 1992). Visuospatial working memory is either reserved or disturbed when the problem is presented vertically to the participants. Finally, we expected approximately similar dependence levels on recall or procedural strategies in several conditions in both high and low math performers. This is based on studies showing common procedural strategies for complex problems (De Smedt et al. 2009; Zhou et al. 2006).

Method

Participants

Eighteen participants (mean age of 21 years and 3 months with a standard deviation of one year and 3 months; one left-handed) were assigned as high math performers and eighteen (mean age of 20 years and 8 months and a standard deviation of one year and 11 months; 2 left-handed) were assigned as low math performers. The participants widely varied in their majors. For the high performers, 13 participants were in the medicine and natural science departments, 2 in management, 2 in media and one in psychology. For the low performers, 6 participants were in the medicine and natural science departments, 4 in art and literature, 3 in education, 3 in law, one in management and one in sport sciences. Participants were assigned in the high or low math performer groups based on a screening test for arithmetic abilities, using a sample of 49 undergraduate female students, aged 18–25 years (mean = 21, standard deviation = 1.6) from King Saud University, recruited via the convenient sampling method. All participants had normal or corrected-to-normal vision with no neurological or sensory disability affecting their task ability. They signed a consent form prior to the experiment and did not receive a reward. The average screening test score was 53.081, with a standard deviation of 18.987. The eighteen participants from each extreme end were assigned as high and low math performers. High math performers scored 54 or above and low math performers scored 45 and below.

Materials

Screening test of arithmetic abilities

This test was a paper-based task of four sections. Each section included one basic arithmetic task type (addition, subtraction, multiplication, and division). The internal consistency, validity and reliability of the test were verified on the current study sample. The correlations between the subsections and the total score ranged between 0.813 and 0.91, $p < 0.001$, and reliability revealed Cronbach's alpha of 0.968. Each section was presented in a single page and included 42 problems. The task time was limited to two minutes for each section (see Online Resource 1 for the detailed description of the task and the included arithmetic problems).

Demography questionnaire

At the beginning of the arithmetic abilities screening test, the participants were asked about their age and major. They were also asked what and when was the last course they studied that involved considerable amount of arithmetic. They were also asked how much of mental arithmetic they use on a daily basis.

Updating task

The updating task was used to assess the participants' ability to actively remove outdated specific information from working memory and substitute it with new incoming information (Ecker et al. 2010; 2014; Kessler and Meiran 2008; Kessler and Oberauer 2014). The current updating task was a computer-based task, following the design of Sidney and colleagues (2018), where participants were presented with a series of letters (12 sequences in total). The sequence length varied between 7 and 12 letters per sequence. Each letter was presented for 1500 ms. The task was to recall the last three letters of each sequence. For example, when a sequence like 'SYMVRKA' appears, participants had to recall the letters 'RKA'. Participants viewed 12 sequences. The first sequence was the same for all participants, with the remaining 11 sequences randomized. The task was corrected by giving one score to each correct item, excluding the first sequence (maximum = 33).

The n-Back task

This computer-based task followed Kaladous and colleagues' design (2015), with minor response method modifications. It had three levels of difficulty (1-back, 2-back, and 3-back). Each level included 40 digits presented sequentially, at the rate of 2,000 ms per stimulus. In the one-back condition, participants had to press the 'ENTER' button if the digit—always a single digit—was the same as the previously presented digits. In the two and three digits, participants had to press 'ENTER' button when the presented digits were presented earlier with a lag of two or three positions, respectively. The n-back task is used to measure a combination of executive functions (Gajewski et al. 2018), including maintenance, updating, inhibiting irrelevant information and matching maintained and new items to initiate a response

(Rac-Lubashevsky and Kessler 2016). The dependent variable was calculated by the d prime equation: $d = z(h) - z(f)$. Z(p) is the reversed cumulative probability function of normal distribution, h is the hits and f is the false alarms (Kaladous et al. 2015).

Digit span

This is a subtest in Wechsler Adult Intelligence Scale (WAIS-I) for adults (Malika 1996). The participants were required to listen to a series of digits and recall the digits in normal order in the first phase of the test and in reverse order in the second phase of the test. The number of items in a series was increased across trials, and participants were stopped after failing to recall the digits in order in two consecutive trials at one level. The serial digit recall assesses auditory verbal short-term memory (STM) span, and inversed recall was assumed to represent the auditory verbal working memory (WM) span.

Coding

This paper-based test is a subtest in WAIS-I for adults (Malika 1996) and required participants to copy symbols assigned to Arabic digits in several lines. It assesses visuospatial short-term memory. The participants were given 60 s instead of 90 s for this task. The score relied on how many symbols were correctly copied in the right boxes (maximum = 67).

Raven's Progressive Matrices

This well-known paper-based test for nonverbal reasoning was administered individually. The version used was validated on the Arabic sample by Ali (2013) and included 36 matrices. The raw score reflected the number of correctly solved matrices.

Right handedness questionnaire

This questionnaire was assembled by Abdul-Kawi (2004) and included 20 items representing daily activities. Participants were required to rate on a 5-point rating scale how much they depended on their right hand for each of the activity.

The mental addition task

A total of 120 addition problems, 60 easy and 60 difficult, were included in the computer-based mental arithmetic task. Generating the problems followed the rules by Tschentscher and Hauk (2015). Easy problems included two single-digit integers, while difficult problems included two two-digit integers. The same number of even and odd operants were used. The number of problems involving carrying were controlled in both conditions. Zero and one were never used in integer units, and neat problems, which have the same units for both operants, were avoided. Restrictions were added so that no total exceeds 100, and the difficulty for all conditions was controlled by controlling the size of totals. As the task was designed to require participants to choose the right answer from two alternatives, one distracter was generated for the solution to each problem. Solutions and distracters were between 1 and 10 units apart (see Online Resource 2 for the addition problems).

Half of the problems were presented horizontally and half vertically. Half of the problems were interleaved between two (5 × 5) visual matrices, where participants had to indicate whether they resembled the first matrix before choosing the right arithmetic problem total. This visuospatial loading task was taken from Cavdaroglu and Knops (2015), with presentation and number of highlighted square modifications. The matrices (5 × 5 cm) were drawn as white squares with green borders. Three squares from each visual matrix were highlighted with red. The location of the red squares was randomly generated for all target matrices. For the test matrices, in which the participants must indicate whether they resemble the target matrices or not, half of the test matrices were the same as targets and half were different.

Brain-computer interface (BCI) for EEG recording

The EPOC + Emotiv BCI is a 16-channel BCI, attached to a computer via Bluetooth. This device records the EEG signals for educational and research purposes. The EEG signals were obtained from the frontal, temporal, occipital regions and parts of the parietal regions of the brain. It included 16 sensors, with two references above the ears, in CMS and DRL. The electrodes were named according to their locations, as adopted by the 10–20 electrode system (Jasper 1958): AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4. The sensors at the end of the electrodes were covered with disposable felt pads designed to be soaked with saline before recording and disposed after each use. This noninvasive device records and sends the EEG to the computer at the rate of 128 samples per second. The EEG signals were amplified via EPOC + Emotiv electrodes and transferred to the EmotivPro software for recording and data processing (Emotiv 2019).

Design and procedures

The design used was a 2 × 2 × 2 × 2 mixed design, with level of math performance (high vs. low math performers) as a between-group variable. Presentation method (horizontal vs. vertical), difficulty (easy vs. difficult) and visuospatial

loading (no load vs. visuospatial load) were within-group variables. The dependent variables were mental addition task accuracy, selected solving strategies and alpha and theta activities in mental addition. Measures of STM and WM (updating, n-back, digit span and coding) and Raven's colored progressive matrices, assessing fluid intelligence, were implemented as controlling variables.

The experiment involved three sessions. In the first session, participants completed the screening test for arithmetic abilities and the demographic questionnaire. This session lasted around 12 min. Following the administration of screening test, 36 participants were chosen and classified as high and low math performers. These 36 participants were asked to attend the second and third sessions of the study.

The second session took place within one to four weeks of the first session and lasted approximately 45 min. The session included six tasks: updating, n-back, digit span, coding, Raven's colored progressive matrices and the right handedness questionnaire. The tests were performed in an individual testing room in a quiet area of the psychology building, which has good lighting conditions and has no windows. All participants performed the tasks in the same order. Each participant first performed the computer-based updating task, then the n-back task on a Dell laptop running Windows 10 with a 15-inch screen and 1,360 × 768 resolution. The tasks ran on e-prime software (v2; Psychology Software Tools, Inc., 2016) and all stimuli were presented in 24 point white font on a black background. Before the n-back task, each participant was given a short practice trial including up to five items to practice the three conditions. After the computer-based tasks, participants performed the digit span test, where the experimenter gave the instructions and an example trial before each of the serial and reverse recall tasks. The experimenter read the participants the digits at the rate of one digit per second. This was followed by the coding test, where the participants were given 8 examples with 60 s to completely copy the codes for the aligned digits. Ultimately, the participants performed the Raven's colored progressive matrices test with a time limit of 15 min. Finally, the participants were asked to complete the righthandedness questionnaire. The participants could take 5-min breaks whenever they desired.

The third session took place two to five weeks after the second session and lasted up to 60 min. This session included the EEG recording while simultaneously performing the computer-based arithmetic task. Participants were seated 60 cm from the laptop screen. At the beginning of this session, each participant was given eight practice trials to practice the horizontal and vertical arithmetic tasks with and without visuospatial load. The BCI headset was connected with EmotivPro software (Emotiv 2019) on a Dell laptop running Windows 10 with a 15-inch screen and a screen resolution of 1,360 × 768. The felt pads were attached to the electrode sensors and put on a participant's head. The time of setting the headset varied between participants. When all electrodes were adjusted, the main mental addition task started on the e-prime software (v2; Psychology Software Tools, Inc., 2016) on the same laptop running EmotivPro.

The 120 addition tasks were presented in four blocks with 30 problems each. The participant was informed of the presentation method and loading conditions at the beginning of each block (horizontal with no load, horizontal with load, vertical with no load, and vertical with load). The participant initiated each block by pressing the spacebar. Each block included 15 easy and 15 difficult problems. The easy and difficult problems were presented continuously in each block and were not mixed. The order of problems was presented differently in a random order for each participant, and the order of blocks was controlled across participants. Each trial was initiated with a fixation cross for 500 ms followed by the addition problem. The font was always 28 point white on a black background. Each arithmetic task was presented in the centre of the screen with an angle view of 4.77 degrees for 5000 ms and was followed by two alternative answers, presented adjacently on a horizontal line and remaining for 1750 ms or until the participant responded by pressing either the right or left arrow key. The right answer was presented on the right half the time and on the left the other half. The time of presentation for the addition task was chosen based on a preliminary study, where the addition task was performed on 16 female undergraduate students. The RT in the preliminary study was ($M = 3547$ ms, $SD = 1648$ ms) for difficult problems and ($M = 2385$, $SD = 1311$) for easy problems. Considering the average RTs and their standard deviations and to decrease the artifacts by finger movement, the time was fixed as 5000 ms, which would be enough for both groups to perform the mental arithmetic tasks without feeling disappointed because of short available time. This did not allow for comparing the RTs, but it helped in selecting similar time windows for analysis of both groups, as will be explained in the results section. Each trial was followed by asking the participants to identify the strategy used to solve the task, either procedural or recall. Participants had to identify their adopted strategy by pressing the right and left arrow keys. For the visuospatial loading task, the 5 × 5 target matrix was presented for 2,000 ms before each addition problem. The test matrix was presented after a participant selected the answer for the arithmetic task and remained for 4000 ms or until the participant responded by pressing the right arrow when the test matrix was the same as the target or the left arrow when it was different. See Fig. 1 for an illustration of the mental addition task. Participants were debriefed after the experiment.

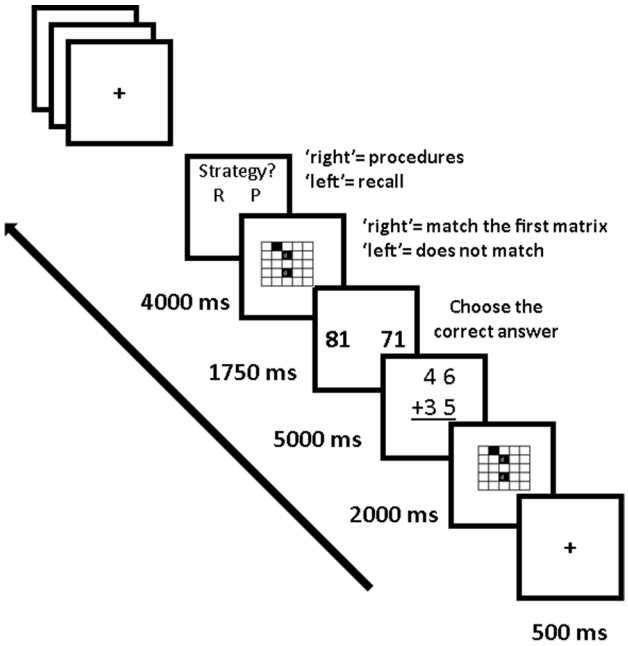

Fig. 1 An illustration of the mental addition task. This is a trial from the loading condition where a visuospatial working memory tasks interleaves around the addition task

Results

Behavioral data

Table 1 shows the scores and independent sample *t* test values for the tests of STM, fluid intelligence, and WM span and processes. High performers significantly surpassed the low performers in STM span, updating, and the 1-back task (see Table 1). Including these variables as covariates in the to-be-presented mixed measures ANOVA on the data of the mental addition task did not reveal different results. Thus, they were not included.

The mental addition task

Performance in the mental addition task was scored by giving one (1) for each correct response and zero (0) for each incorrect response (maximum = 120). Percentages of correct responses were then calculated. A mixed measures ANOVA with level of performance (high vs. low performers) as a between group factor and presentation method (horizontal vs. vertical), loading (no load vs. with load), and difficulty (easy vs. difficult) was performed to examine said variables' effects on mental addition accuracy. The test revealed the main effect of performance level, $F(1, 34) = 46.717$, $p < 0.001$, $\eta^2_p = 0.579$, showing the expected advantage among high math performers ($M = 90.323$, $SD = 3.106$) over low math performers ($M = 75.971$, $SD = 8.409$). It also showed a significant effect of difficulty, $F(1, 34) = 147.271$, $p < 0.001$, $\eta^2_p = 0.812$, where mean performance in easy problems ($M = 92.222$, $SD = 7.4$) is significantly higher than performance in difficult problems ($M = 74.073$, $SD = 13.768$) across all settings and groups. There was also a significant interaction between difficulty and level of performance, $F(1, 34) = 19.325$, $p < 0.001$, $\eta^2_p = 0.362$. To explore the significant interaction between performance level and difficulty, two paired-sample *t* tests compared the effect of performance level on mental addition accuracy within each of easy and difficult problems. The *p* value, here and in all multiple comparisons, was corrected using the Bonferroni correction method. There was found to be a significant difference between high and low performers in the easy problems, $t(34) = 3.673$, $p = 0.003$, and a significant difference in the difficult problems, $t(34) = 7.052$, $p < 0.001$. However, the mean of the differences between high and low math performers in the

Table 1 Means and standard deviations for the controlling tasks for high and low math performers

	High math performers		Low math performers		Independent sample *t* test	
	M	*SD*	*M*	*SD*	*t*(34)	*p*
Screening of math ability	68.555	13.556	34.777	7.042	9.831	.001**
Digit updating	24.611	6.184	19.666	7.829	2.103	.043*
1-back	3.725	0.285	3.170	0.982	2.300	.028*
2-back	2.650	0.557	2.295	0.742	1.626	.113
3-back	1.832	0.413	1.635	0.463	1.347	.187
Digit span (serial recall)	5.833	0.707	5.222	0.427	3.137	.004**
Digit span (reversed)	4.500	0.857	4.000	0.766	1.844	.074
Coding	26.972	5.489	23.111	6.346	1.952	.059
Raven's matrices	33.000	3.086	32.555	2.406	0.482	.633

*$p < .05$, **$p < .01$

easy conditions ($M = 8.518$, $SD = 8.496$) was smaller than that in the difficult problems ($M = 20.925$, $SD = 12.089$), $t(17) = 4.455$, $p < 0.001$.

Another interaction was found between presentation method and loading $F(1, 34) = 5.592$, $p = 0.024$, $\eta^2 p = 0.141$. Neither presentation method nor loading had a significant main effect $F(1, 34) = 0.091$, $p = 0.764$; and $F(1, 34) = 1.231$, $p = 0.275$, respectively. To explore the significant interaction between presentation method and loading, two paired-sample t tests compared the effect of loading vs. no loading on mental addition accuracy within each of the horizontally- and vertically-presented problems. There was found to be a difference between loading and no loading in the horizontally-presented problems, $t(35) = 2.173$, $p = 0.018$, while no difference was found in the vertically-presented problems, $t(35) = 0.771$, $p = 0.223$. See Fig. 2 for the means and standard errors of accuracy in horizontal and vertical conditions.

The visuospatial working memory task

For participant performance in the visuospatial working memory task, the scoring criterion was one (1) for correct and zero (0) for incorrect responses, and then percentages were calculated. See Fig. 3 for the means and standard errors of accuracy in performance in the visuospatial working memory task. A mixed measures ANOVA with level of performance as a between group variable and presentation method and difficulty as within group variables was performed to test whether performance in the visuospatial working memory secondary task varied under different conditions. All main effects were insignificant: level of performance, $F(1, 34) = 2.621$, $p = 0.115$; presentation method, $F(1, 34) = 2.933$, $p = 0.096$; and difficulty, $F(1, 34) = 1.036$, $p = 0.316$. A significant interaction was found between presentation method and difficulty, $F(1, 34) = 6.995$, $p = 0.012$, $\eta^2 p = 0.171$. This interaction is explained by an unexpected significant improvement of performance in visuospatial working memory when performed with horizontal difficult arithmetic

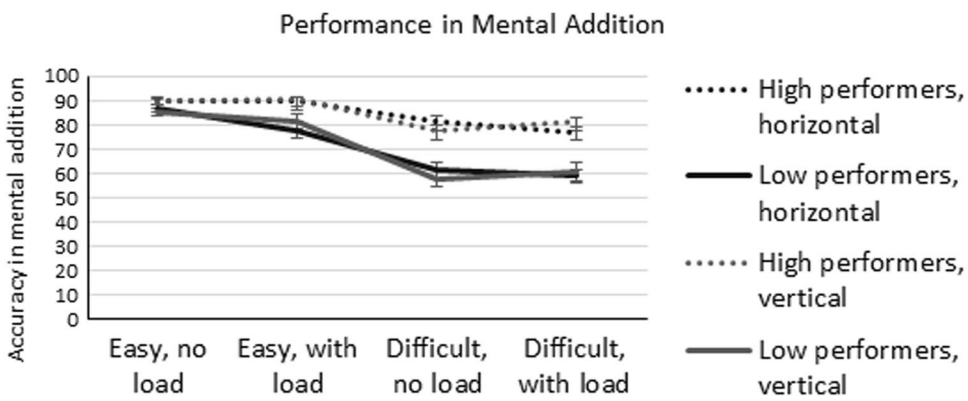

Fig. 2 The mean accuracy in mental addition in the horizontally and vertically presented problems. Error bars represent standard errors

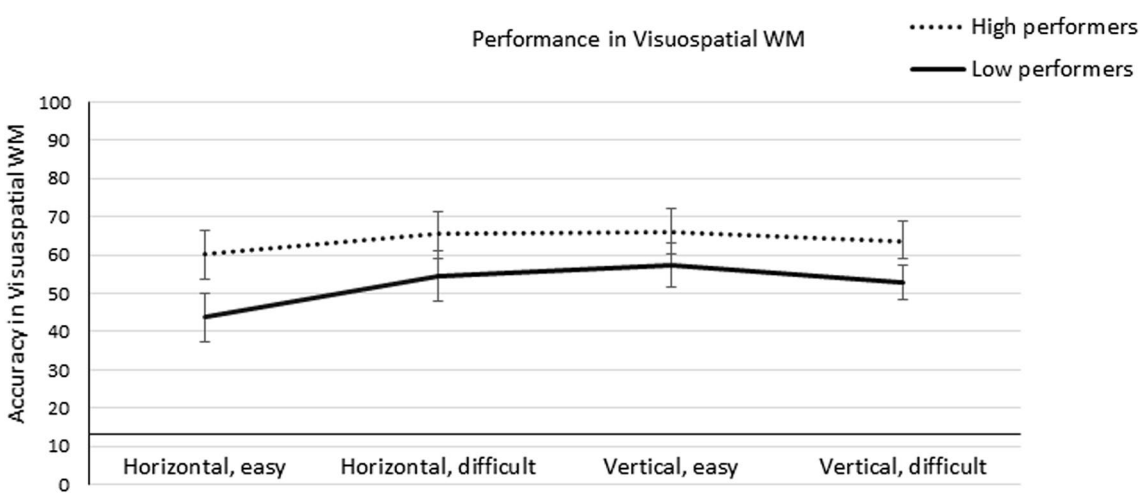

Fig. 3 The mean accuracy in performance in the secondary visuospatial working memory task. Error bars represent standard errors

problems ($M = 64.073$, $SD = 27.793$) compared to horizontal easy problems ($M = 55.366$, $SD = 29.333$), $t(34) = 2.560$, $p = 0.015$. Performance in visuospatial working memory in the easy ($M = 65.74$, $SD = 27.046$) and difficult ($M = 62.033$, $SD = 22.413$) vertically-presented problems was not significantly different, $t(34) = 1.111$, $p = 0.274$.

Solving strategies

It was aimed to examine whether high and low performers used different strategies based on task demands related to presentation method, loading of visuospatial memory, and task difficulty. Therefore, the number of trials where the participant selected the procedural strategies was counted for each participant in each condition (maximum was 15). Table 2 shows the means of the percentages of problems that recruited procedural strategies as well as the standard deviations for each condition. A mixed measures ANOVA was performed on the number of problems, with procedural strategies in each condition as the dependent variable, level of performance as a between group variable, and presentation method, loading, and difficulty as within group variables. Paired-sample *t* tests were used to follow up significant interactions. No significant difference was found between the two groups in terms of performance level, $F(1, 34) = 1.686$, $p = 0.203$. The main effect of presentation method was likewise insignificant, $F(1, 34) = 1.178$, $p = 0.285$.

However, significant main effects were found for both loading, $F(1, 34) = 37.204$, $p < 0.001$, $\eta^2 p = 0.522$, and difficulty, $F(1, 34) = 83.807$, $p < 0.001$, $\eta^2 p = 0.711$, showing a greater use of procedural strategies for non-loading conditions and for difficult problems. There was a significant interaction between presentation method and level of performance, $F(1, 34) = 4.616$, $p = 0.039$, $\eta^2 p = 0.120$. High performers showed a greater use of procedural strategies in vertically than in horizontally presented problems, $t(17) = 2.455$, $p = 0.025$, while low performers showed a similar average use of procedural strategies in both vertically and horizontally presented problems, $t(17) = 0.706$, $p = 0.489$. Additionally, a significant interaction was found between difficulty and level of performance, $F(1, 34) = 5.035$, $p = 0.031$, $\eta^2_p = 0.129$. Low performers showed a significantly greater use of procedural strategies for easy problems ($M = 23.222$, $SD = 12.056$) compared to high performers ($M = 15.722$, $SD = 8.1$), $t(34) = 2.110$, $p = 0.042$. In contrast, both low performers ($M = 32.888$, $SD = 12.915$) and high performers ($M = 31.666$, $SD = 9.081$) showed similar percentages in terms of use of procedural strategies for difficult problems, $t(34) = 0.328$, $p = 0.745$.

Table 2 Means and standard deviations of percentages of addition problems, which used procedural strategies in each experimental condition in the mental addition task

Method of presentation	Loading and difficulty	High math performers		Low math performers	
		M	*SD*	*M*	*SD*
Horizontal	No load, easy	21.846	18.686	46.293	26.113
	No load, difficult	70	23.786	72.586	20.433
	With load, easy	29.253	22.646	33.333	27.626
	With load, difficult	23.333	27.653	39.253	37.14
Vertical	No load, easy	25.553	16.206	46.293	27
	No load, difficult	87.033	16.873	70.74	23.774
	With load, easy	28.146	26.226	28.886	34.973
	With load, difficult	30.74	29.026	36.666	34.54

Electrophysiological data

As discussed in the method section, continuous EEG was recorded using EPOC + BCI (Emotiv, US) with 14 electrodes positioned according to the 10–20 system (Jasper 1958) on the participants' scalp. The 14 electrodes were AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4. Analyses were performed offline using 2019b version of MATLAB. During pre-processing, the EEG was band-pass filtered from 0.5 to 45, digitized at a sampling rate of 128 Hz. To eliminate power line interferences, 50 Hz and 60 Hz notch filters were implemented. Epochs from 500 ms pre-stimulus to 2500 ms post-stimulus were extracted from continuous EEG signals. This time window was chosen because the average participant RT in the preliminary study indicated that this period was occupied by arithmetic problem mental processing. Epochs were visually inspected, and artifact rejection based on the mean signal amplitude during the epoch period was separately performed for each epoch. To reduce the possibility of edge artifacts, modified periodogram was applied to the signal tapered by a rectangular window. Spectral power was computed for every frequency band. This procedure was performed for each epoch, electrode, and participant. Participants with less than 80 intact epochs were removed from further analysis; data from five participants (four low and one high math performers) were removed due to this criterion. Finally, frequency bands corresponding to alpha (8–12 Hz) and theta (4–7.5 Hz) bands were extracted. The mean power for each band was calculated, and the alpha and theta band signals were extracted by applying discrete wavelet transform.

Group analysis included the channels related to the prefrontal areas (AF3 and AF4), frontal and central frontal areas (F3, F4, FC5, FC6), parietal lobe (P7 and P8), and occipital lobe (O1 and O2). Further analysis excluded the temporal lobe, since previous literature contains no evidence of the temporal lobe's involvement in mental arithmetic.

An analysis was first performed across all channels in each hemisphere to inspect the main activities. A mixed measures ANOVA with level of performance as a between group variable, and channels, presentation method, loading, and difficulty as within group variables was performed for each alpha and theta in each hemisphere. For alpha band activities in the left hemisphere, the results showed a significant main channel effect, $F(4, 116) = 5.118$, $p = 0.001$, $\eta^2_p = 0.150$. For theta band activities, analysis across all channels in the left hemisphere showed a significant interaction effect between channel, loading, and difficulty, $F(4, 116) = 3.123$, $p = 0.018$, $\eta^2_p = 0.097$, and a significant main channel effect, $F(4, 116) = 5.782$, $p < 0.001$, $\eta^2_p = 0.166$.

Based on the above analysis and considering that data of mental addition only showed main effects for performance level and difficulty, the analysis of EEG data focused on these two variables. The remaining variables (method of presentation and loading) were collapsed together (i.e., epoch signals belonging to the same condition: level of performance (high vs. low performers) x difficulty (easy vs. difficult) were averaged). Figures 4 and 5 show the topographical images of alpha and theta activities when grand means

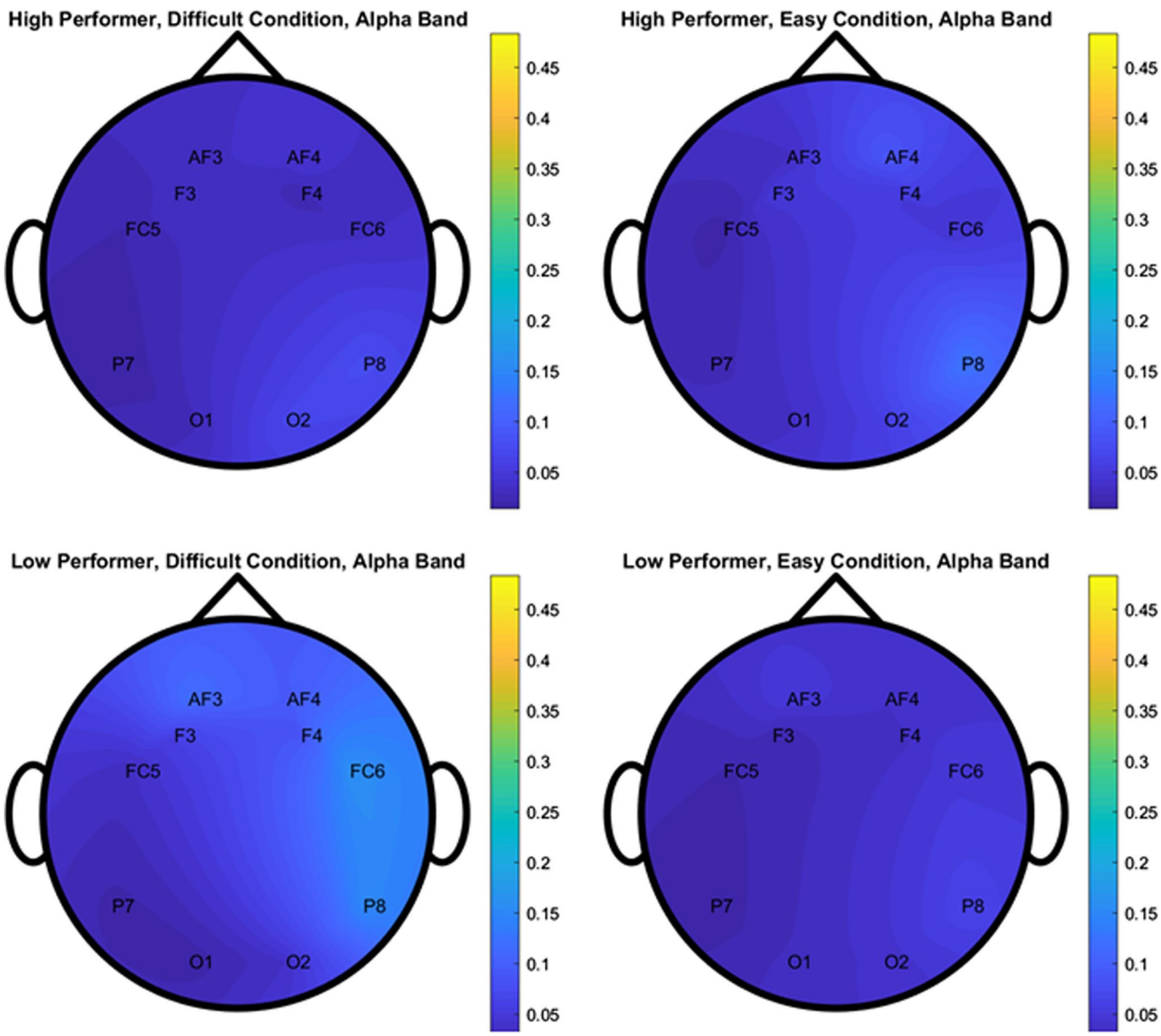

Fig. 4 Topographical images of alpha power in the cortex. Values of alpha power are the grand means for all epochs in each condition, depicted for each image. As indicated by the labels of each image, the grand means are generated after collapsing all easy conditions in one grand mean and all difficult conditions in another grand mean for each of the low and high math performers groups. The right bar for each image indicates the values of the color codes in the topographical image. In P7, the value of alpha activities in low performers was smaller than in high performers in both easy and difficult conditions

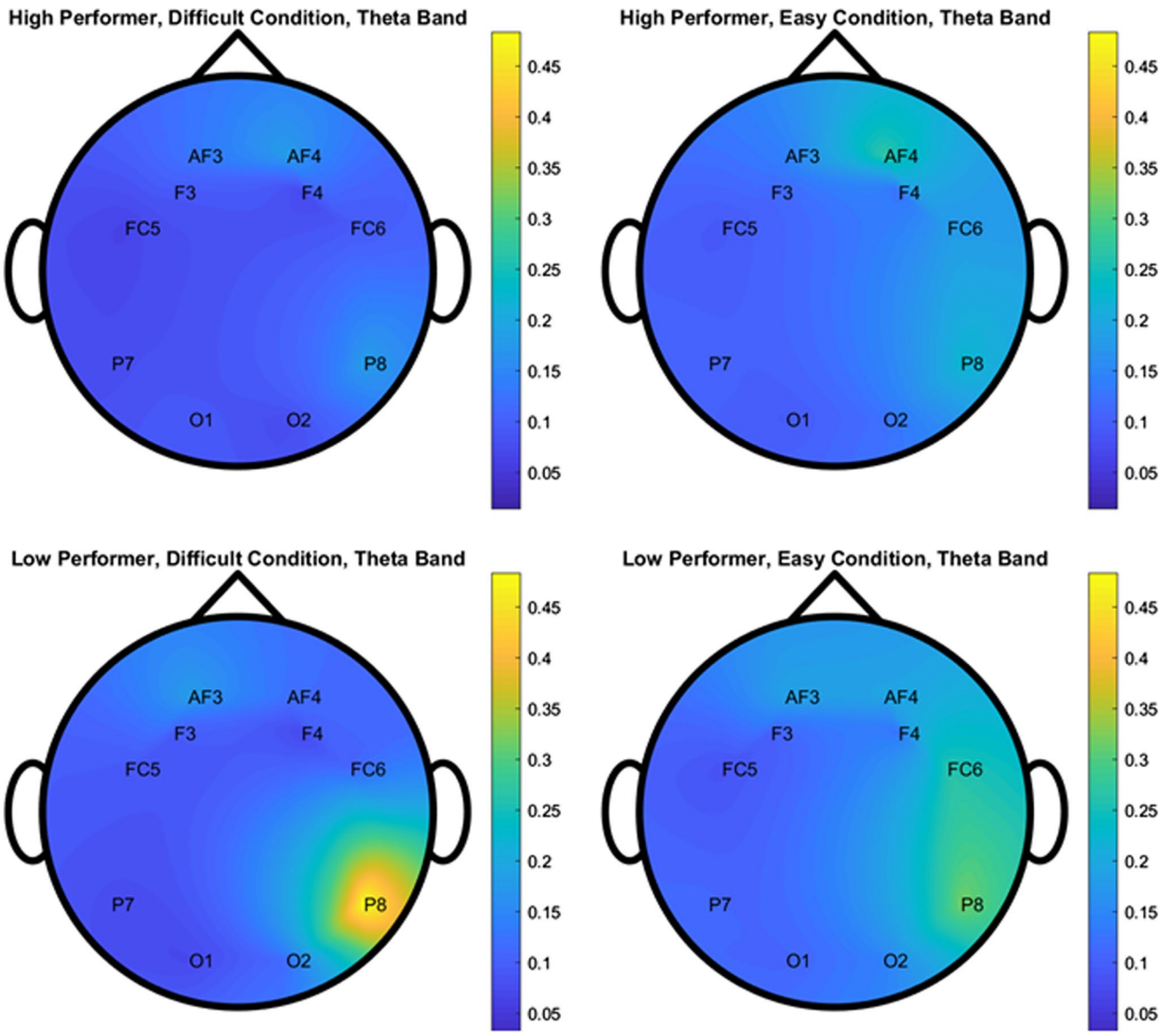

Fig. 5 Topographical images of theta power in the cortex. Values of theta powers are the grand means for all epochs in each condition, depicted for each image. As indicated by the labels of each image, the grand means are generated after collapsing all easy conditions in one grand mean and all difficult conditions in another grand mean for each of the low and high math performers groups. The right bar for each image indicates the values of the color codes in the topographical image. Changes in theta activities according to math performance and difficulty was not significant

are collapsed across easy and difficult conditions for each of the high and low performer groups. A mixed measures ANOVA was then performed for each channel with level of performance (high vs. low math performers) as a between group variable, and difficulty as a within group variable. The p values were corrected via Bonferroni correction method. This analysis was performed for each of the alpha and theta activities. Tables 3 and 4 show the results of mixed measures ANOVA for each channel for alpha and theta, respectively. Alpha power significantly differed between individuals with high and low math performance at the left parietal site (P7), indicating a larger decrease in alpha power for individuals with low as compared to high math performance, $F(4, 116) = 14.139$, $p < 0.001$, $\eta^2_p = 0.329$.

Discussion

This study explored the individual differences in mental arithmetic by comparing the accuracy of high and low math performers when performing mental addition in different settings. The related brain oscillations during mental addition,

Table 3 Alpha activities during mental addition, calculated by mixed-measures ANOVA

Hemisphere	Channel	Variable	$F(1, 29)$	p	η^2p
Left	AF3	Difficulty	5.599	.025	.162
		Level of performance	0.667	.421	.022
		Interaction	0.483	.492	.016
	F3	Difficulty	2.378	.134	.076
		Level of performance	1.055	.313	.035
		Interaction	0.737	.398	.025
	FC5	Difficulty	1.732	.198	.056
		Level of performance	0.001	.979	.000
		Interaction	0.016	.900	.001
	P7	Difficulty	1.126	.279	.040
		Level of performance	14.193	.001**	.329
		Interaction	1.829	.187	.059
	O1	Difficulty	0.003	.955	.000
		Level of performance	2.764	.107	.087
		Interaction	0.019	.891	.001
Right	AF4	Difficulty	0.746	.395	.025
		Level of performance	4.317	.047	.130
		Interaction	0.272	.606	.009
	F4	Difficulty	4.130	.051	.125
		Level of performance	0.855	.363	.029
		Interaction	0.006	.941	.001
	FC6	Difficulty	0.308	.583	.010
		Level of performance	0.052	.821	.002
		Interaction	0.312	.581	.011
	P8	Difficulty	1.923	.176	.062
		Level of performance	0.450	.508	.015
		Interaction	0.822	.372	.028
	O2	Difficulty	0.157	.695	.005
		Level of performance	4.110	.052	.124
		Interaction	0.106	.747	.004

Bonferroni alpha level is corrected to .01. (**) = $p < .01$

Table 4 Theta activities during mental addition, calculated by mixed-measures ANOVA

Hemisphere	Channel	Variable	$F(1, 29)$	p	η^2p
Left	AF3	Difficulty	2.832	.103	.089
		Level of performance	0.143	.709	.005
		Interaction	0.010	.921	.000
	F3	Difficulty	1.600	.216	.052
		Level of performance	1.599	.216	.052
		Interaction	0.000	.990	.000
	FC5	Difficulty	2.046	.163	.066
		Level of performance	0.410	.527	.014
		Interaction	0.034	.855	.001
	P7	Difficulty	0.016	.901	.001
		Level of performance	1.074	.309	.036
		Interaction	0.733	.399	.025
	O1	Difficulty	1.561	.222	.051
		Level of performance	3.476	.072	.107
		Interaction	0.040	.843	.001
Right	AF4	Difficulty	0.354	.557	.012
		Level of performance	3.410	.075	.105
		Interaction	0.059	.810	.002
	F4	Difficulty	2.904	.099	.091
		Level of performance	1.098	.303	.036
		Interaction	0.454	.506	.015
	FC6	Difficulty	0.223	.640	.008
		Level of performance	0.779	.385	.026
		Interaction	0.104	.749	.004
	P8	Difficulty	4.176	.050	.126
		Level of performance	0.360	.553	.012
		Interaction	2.245	.145	.072
	O2	Difficulty	0.647	.428	.022
		Level of performance	2.886	.100	.090
		Interaction	0.079	.780	.003

Bonferroni alpha level is corrected to .01.

specifically alpha and theta oscillations, were recorded and analyzed. The results demonstrate an interaction between the participants' mental addition performance level and the problem difficulty levels. The gap between high and low performers increased with difficult problems. Both groups of participants seemed to use procedural strategies more frequently when faced with difficult addition problems and when no visuospatial load interrupted their visuospatial working memory. Regarding the electrophysiological data, the effect of problem difficulty on alpha and theta power was examined for each channel. In the left posterior parietal region, recorded by the P7 electrode, alpha power showed a larger decrease for the low than the high math performer group. These findings support the view that differences in performance can be detected using behavioral and neural-based measures (Núñez-Peña et al. 2011).

The most significant finding is the difference between high and low math performers in P7, whereby alpha power showed a significant decrease for low math performers, in contrast to high math performers. Although the behavioral data showed greater mental addition accuracy differences between high and low performers in difficult compared to easy conditions, low performers' alpha showed an overall decrease for both easy and difficult conditions. The decrease

in alpha in the parietooccipital area was found to be related to the use of procedural strategies in mental arithmetic (De Smedt et al. 2009). Furthermore, working memory, which is required for procedural strategies, is related to a decrease in alpha in several regions, including the posterior parietal region (Borghini et al. 2014; Foxe and Snyder 2011; Klimesch 1999).

The current study showed an interaction between the level of performance and the effect of difficulty on using procedural strategies, whereby the post hoc analysis indicated a greater use of procedural strategies by low performers compared to high performers under easy conditions. This is consistent with the finding of decreased alpha power under easy conditions for low performers. For the difficult conditions, not only did both high and low performers show a similar level of dependence on procedural strategies under difficult conditions, but also the high performers showed a larger dependence on procedural strategies in the vertical condition. It might be surprising, at first glance, that low performers, rather than high performers, showed a decrease in alpha power under the difficult condition. Nevertheless, despite the reliance on procedural strategies, high performers automatically retrieve intermediate answers, and this ability is not available for low performers (Jackson and Coney 2007). Low performers depend on procedural strategies to a larger extent, even for many easy problems and intermediate answers. Therefore, the decrease in alpha in low performers emerges under both easy and difficult conditions.

In contrast, high performers depend heavily on the retrieval of answers in easy problems and use retrieval in conjunction with procedural strategies as a tool to retrieve intermediate answers. This claim receives support from previous studies. High activation in the left angular gyrus, responsible for arithmetic fact retrieval, was found in high math performers regardless of the task (Grabner et al. 2007). High math performers were also previously found to depend more on retrieval strategies in mental arithmetic (Grabner et al. 2007, 2009; Thevenot 2007).

Another possible explanation for the decrease in alpha activities in low math performers in both easy and difficult problems relates to the effect of math anxiety. A previous study found that cortical activities increased in participants with high math anxiety when they were given arithmetic problems, regardless of the difficulty of the task (Klados et al. 2015). Math anxiety is related to a reduction in math performance (Sidney et al. 2018; Trezise and Reeve 2014) as it consumes working memory resources (Ashkenazi and Najjar 2018; Trezise and Reeve 2014) or causes the avoidance of math problems (Ashcraft and Kirk 2001). For a review of the disrupting effect of math anxiety on math performance, see Ashcraft and Kirk (2001).

It is interesting to note that the decrease in alpha power occurs only in the left posterior parietal lobe. According to the triple-code model of Dehaene (1992), the posterior parietal lobe is related to spatial and non-spatial attention, the left angular gyrus is related to the recall of verbal information, and the intraparietal sulcus is related to number magnitudes and the representation of the number line. As discussed, the triple-code model has provided evidence that attention is required for procedural strategies as they involve multiple processes to reach the final answer to each problem. Procedural strategies in mental addition seem to exhibit a large use of visuospatial attention, which is related to the left posterior parietal lobe (Zago et al. 2001); hence, the decrease of alpha is significant only in the left posterior parietal region.

This interpretation is consistent with findings from previous studies that showed a left-side dominance in mental arithmetic in both high and low performers (Proverbio et al. 2019) and for procedural strategies (Kong et al. 2005). In previous studies, the left and right regions in the parietal lobes showed different patterns of activity during mental arithmetic (Delazer et al. 2005; Prado et al. 2013).

However, it should also be mentioned that in previous studies, bilateral posterior parietal regions were found to show a decrease in alpha power with difficult mental arithmetic (De Smedt et al. 2009). Additionally, fMRI findings showed more activities in biparietoccipital areas with complex tasks (Menon et al. 2000b) and procedural strategies (Grabner et al. 2009). The role of the left posterior parietal lobe is related to visuospatial attention in procedural strategies, and further studies can explore the specific roles of the right posterior parietal lobe in procedural strategies.

The findings of this study show that in the left hemisphere, problem difficulty and the loading of visuospatial working memory have different effects on theta power in different areas. However, the post hoc analysis failed to show any significant results. Theta is responsible for memory encoding (Klimesch 1999) and executive processing (Sausing et al. 2005), and it has been found in anterior frontoparietal regions in deductive reasoning tasks, which are highly demanding tasks. Additionally, an increase in theta power is related to the retrieval of arithmetic facts (De Smedt et al. 2009). However, as the study found no significant findings in relation to this, the role of theta will not be further discussed here.

With regard to solving strategies, participants' more frequent use of procedural strategies with no visuospatial load illustrates that the visuospatial sketchpad is critical for procedural strategies. This is consistent with previous findings, where visuospatial loading disrupted performance in procedural strategies (Hubber et al. 2013).

Another interesting finding is that visuospatial loading impaired performance in horizontally presented arithmetic problems, yet it had no significant effect on vertically presented problems. This interfering effect of loading

visuospatial working memory on horizontally presented arithmetic problems is complicated by the finding that the performance in the secondary visuospatial working memory task was higher when performed concurrently with difficult, instead of easy, horizontally presented arithmetic problems. This indicates that the participants either focus on the visuospatial working memory task or the mental addition task. Furthermore, this emphasizes the role of visuospatial working memory in mental arithmetic and shows that both high and low performers might rely less on visuospatial working memory when arithmetic problems are arranged in a vertical order. Dowker (2005) concluded that flexibility in strategy usage is required for accurate mental arithmetic performance. In this study, the high math performers tended to use procedural strategies more with vertically, than with horizontally, presented problems, meaning they were more flexible in changing their strategy according to how the addition problems were presented.

A previous study on Chinese students found that visuospatial loading impairs vertically presented problems, but not horizontal ones (Imbo and LeFevre 2010). However, in this study, visuospatial loading impairment was found on horizontal, and not vertical, problems. Future studies can explore the role of the visuospatial sketchpad and related occipital lobe activities in arithmetic in connection with cultural and individual differences.

A final note is related to the differences between high and low math performers in the current study. As stated in the results, high and low math performers, categorized based on their performance in an arithmetic fluency test, also differ in their verbal STM and updating abilities, which were assessed by the digit span task and 1-back task, respectively. N-back measures a combination of executive functions, including updating (Gajewski et al. 2018). The 1-back task is less demanding than the 2- and 3-back versions. High and low math performers only differ in their performance on the least demanding task. This was previously explained by the activation of spatial representation (Fischer et al. 2003) or number semantic representation via number stimuli (Knops et al. 2006). These semantic representations of numbers are activated and enhance performance when faced with simple tasks. When the load on working memory increases (e.g., under 2- and 3-back conditions), the effect of number semantic representations interferes with working memory. The difference between high and low performers in number representation might be the reason for the differences between the two groups in the 1-back task and in digit serial recall.

It could be argued that using number stimuli to assess working memory activates other semantic and spatial representations of numbers, which are components of mental arithmetic (Knops et al. 2006). Nevertheless, low math performers are behind in both their digit-serial recall and n-back, which use number stimuli, and working memory updating, which uses letters. Therefore, the differences in working memory are not solely attributable to differences in number representation. Furthermore, it is hard to disassociate working memory efficiency from mental arithmetic skills. The order of items in working memory activates magnitude representations that are similar to number representations (Marshuetz et al. 2000). Updating involves multiple component processes: retrieval, transformation, and substitution (Ecker et al. 2010, 2014; Kessler and Meiran 2008). In mental arithmetic, updating seems to be particularly necessary when carrying one and two-digit integers, whereby participants update decades alongside retrieving the unit total. Updating processes are the core processes in working memory (Kessler and Oberauer 2014), which, in turn, is the executor of arithmetic calculations that do not depend on mere solution retrieval (Dowker 2005).

Given the small number of participants in this study, caution must be exercised in generalizing the findings and interpretations. Another limitation is that the study included only female participants. Evidence from previous findings suggests a different utilization of solving strategies based on gender (Pletzer 2016). Hence, replications and extensions of the current findings to both males and females are required.

Conclusion

High vs. low math performers respond differently to arithmetic problems in both behavioral and neural measures. They use different strategies in easy problems and their accuracies in mental addition are significantly different. Furthermore, while they tend to use similar strategies for difficult problems, the efficiency of low performers is not at the same level as that of high performers, which leads to significantly lower accuracy. The current study showed that a decrease in alpha, which indicates cortical arousal, in the posterior parietal region is related to lower mental arithmetic skills. Combining behavioral and neural measures is a promising approach for understanding differences in math performance.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1007/s10339-021-01038-1.

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Data availability This study's data is available from the author upon request.

Declarations

Conflict of interest The author declares there is no conflict of interest in publishing the research results.

Ethical approval All procedures performed involving human participants were in accordance with the institutional research committee's ethical standards [The deanship of Scientific Research, King Saud University: KSU-KSU-HE-19–426], and with the 1964 Helsinki declaration and its later amendments.

Informed consent Informed consent was obtained from all participants in the study.

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